

A decorative graphic on the left side of the slide, consisting of a black crosshair overlaid on a blue square, a red square, and a yellow square.

Lecture 2

Chapter 1. The Foundations

1.1 Propositional Logic

Review: The Implication Operator



- The conditional statement (a.k.a. ***implication***) $p \rightarrow q$ states that p implies q .
- */i.e., If p is true, then q is true; but if p is not true, then q could be either true or false.*
- *E.g., let p = “You study hard.”
 q = “You will get a good grade.”
 $p \rightarrow q$ = “If you study hard, then you will get a good grade.” (else, it could go either way)*
 - *p : hypothesis or antecedent or premise*
 - *q : conclusion or consequence*

Review: Implication Truth Table



p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

The only
False case!

- $p \rightarrow q$ is **false** only when p is true but q is **not** true.
- $p \rightarrow q$ does **not** require that p or q are ever true!
 - E.g. “ $(1=0) \rightarrow$ pigs can fly” is TRUE!

Examples of Implications

- “If this lecture ever ends, then the sun will rise tomorrow.” *True or False?* ($T \rightarrow T$)
- “If $1+1=6$, then Obama is president.” *True or False?* ($F \rightarrow T$)
- “If the moon is made of green cheese, then I am richer than Bill Gates.” *True or False?* ($F \rightarrow F$)
- “If Tuesday is a day of the week, then I am a penguin.” *True or False?* ($T \rightarrow F$)

English Phrases Meaning $p \rightarrow q$

- “ p implies q ”
- “if p , then q ”
- “if p , q ”
- “when p , q ”
- “whenever p , q ”
- “ q if p ”
- “ q when p ”
- “ q whenever p ”
- “ p only if q ”
- “ p is sufficient for q ”
- “ q is necessary for p ”
- “ q follows from p ”
- “ q is implied by p ”

We will see some equivalent logic expressions later.

Converse, Inverse, Contrapositive

- Some terminology, for an implication $p \rightarrow q$:
- Its **converse** is: $q \rightarrow p$.
- Its **inverse** is: $\neg p \rightarrow \neg q$.
- Its **contrapositive**: $\neg q \rightarrow \neg p$.

p	q	$p \rightarrow q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$	$\neg q \rightarrow \neg p$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

- One of these three has the *same meaning* (same truth table) as $p \rightarrow q$. Can you figure out which?

Contrapositive

Examples

- p : Today is Easter
 q : Tomorrow is Monday
- $p \rightarrow q$:
If today is Easter then tomorrow is Monday.
- **Converse**: $q \rightarrow p$
If tomorrow is Monday then today is Easter.
- **Inverse**: $\neg p \rightarrow \neg q$
If today is not Easter then tomorrow is not Monday.
- **Contrapositive**: $\neg q \rightarrow \neg p$
If tomorrow is not Monday then today is not Easter.

The Biconditional Operator

- The ***biconditional*** statement $p \leftrightarrow q$ states that p ***if and only if*** (*iff*) q .
- $p =$ “It is below freezing.”
 $q =$ “It is snowing.”
 $p \leftrightarrow q =$ “It is below freezing if and only if it is snowing.”

or

$=$ “That it is below freezing is necessary and sufficient for it to be snowing”

Biconditional Truth Table

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

- p is necessary and sufficient for q
- If p then q , and conversely
- p iff q

- $p \leftrightarrow q$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$.
- $p \leftrightarrow q$ means that p and q have the **same** truth value.
- $p \leftrightarrow q$ does **not** imply that p and q are true.
- Note this truth table is the exact **opposite** of $\oplus$'s!
Thus, $p \leftrightarrow q$ means $\neg(p \oplus q)$.

Boolean Operations Summary

- Conjunction: $p \wedge q$, (read p and q), “discrete math is a required course **and** I am a computer science major”.
- Disjunction: $p \vee q$, (read p or q), “discrete math is a required course **or** I am a computer science major”.
- Exclusive or: $p \oplus q$, “discrete math is a required course **or** I am a computer science major **but not both**”.
- Implication: $p \rightarrow q$, “**if** discrete math is a required course **then** I am a computer science major”.
- Biconditional: $p \leftrightarrow q$, “discrete math is a required course **if and only if** I am a computer science major”.

Boolean Operations Summary

- We have seen 1 unary operator and 5 binary operators. What are they? Their truth tables are below.

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	F	T	T
T	F	F	F	T	T	F	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

- For an implication $p \rightarrow q$
- Its **converse** is: $q \rightarrow p$
- Its **inverse** is: $\neg p \rightarrow \neg q$
- Its **contrapositive**: $\neg q \rightarrow \neg p$

Compound Propositions

- A **propositional variable** is a variable such as p , q , r (possibly subscripted, e.g. p_i) over the Boolean domain.
- An **atomic proposition** is either Boolean constant or a propositional variable: e.g. T , F , p
- A **compound proposition** is derived from atomic propositions by application of propositional operators: e.g. $\neg p$, $p \vee q$, $(p \vee \neg q) \rightarrow q$
- Precedence of logical operators: $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$
- Precedence also can be indicated by parentheses.
 - e.g. $\neg p \wedge q$ means $(\neg p) \wedge q$, not $\neg(p \wedge q)$

An Exercise

- Any compound proposition can be evaluated by a truth table
- $(p \vee \neg q) \rightarrow q$

p	q	$\neg q$	$p \vee \neg q$	$(p \vee \neg q) \rightarrow q$
T	T	F	T	T
T	F	T	T	F
F	T	F	F	T
F	F	T	T	F

Translating English Sentences

- Let p = “It rained last night”,
 q = “The sprinklers came on last night,”
 r = “The lawn was wet this morning.”

Translate each of the following into English:

$$\neg p = \text{“It didn’t rain last night.”}$$

$$r \wedge \neg p = \text{“The lawn was wet this morning, and it didn’t rain last night.”}$$

$$\neg r \vee p \vee q = \text{“The lawn wasn’t wet this morning, or it rained last night, or the sprinklers came on last night.”}$$

Another Example

- Find the converse of the following statement.
 - “Raining tomorrow is a sufficient condition for my not going to town.”
- **Step 1:** Assign propositional variables to component propositions.
 - p : It will rain tomorrow
 - q : I will not go to town
- **Step 2:** Symbolize the assertion: $p \rightarrow q$
- **Step 3:** Symbolize the converse: $q \rightarrow p$
- **Step 4:** Convert the symbols back into words.
 - “If I don’t go to town then it will rain tomorrow” or
 - “Raining tomorrow is a *necessary condition* for my not going to town.”

Logic and Bit Operations

- A **bit** is a **b**inary (base 2) dig**it**: 0 or 1.
- Bits may be used to represent truth values.
 - By convention:
0 represents “False”; 1 represents “True”.
- A **bit string of length n** is an ordered sequence of $n \geq 0$ bits.
- By convention, bit strings are (sometimes) written left to right:
 - e.g. the “first” bit of the bit string “1001101010” is 1.
 - What is the length of the above bit string?

Bitwise Operations

- Boolean operations can be extended to operate on bit strings as well as single bits.

- Example:

01 1011 0110

11 0001 1101

11 1011 1111 Bit-wise OR

01 0001 0100 Bit-wise AND

10 1010 1011 Bit-wise XOR

End of 1.1

You have learned about:

- Propositions: what they are
- Propositional logic operators'
 - symbolic notations, truth tables, English equivalents, logical meaning
- Atomic vs. compound propositions
- Bits, bit strings, and bit operations
- Next section:
 - Propositional equivalences
 - Equivalence laws
 - Proving propositional equivalences